SUCCESSFUL REPAIR OF ESOPHAGEAL INJURY USING ELASTIN BASED BIOMATERIAL PATCH

Michio Kajitani MD, PhD, Yasmin Wadia MD, Monica T. Hinds PhD, Jeffrey Teach RN, Kim Swartz MD, Kenton G. Gregory MD
Oregon Medical Laser Center, 9205 SW Barnes Rd. Portland, OR 97225

Purpose: An esophageal injury with significant tissue loss is very difficult to repair primarily, and has high morbidity and mortality with stricture and leak. We conducted in vivo study to test our elstin based biomaterial patch composed of elastin and collagen to repair the large tissue defect in the esophagus.

Method: The elastin based biomaterial was made from porcine aorta, by decellulization, and sterilization. Collagen coating on the adventitia was preserved to retain mechanical strength and to enhance cellular ingrowth. Ten domestic pigs underwent right thoracotomy. A 2cm circular defect was made on the distal esophagus, excising half of its circumference, and was repaired using the elastin based biomaterial patch and running absorbable sutures. Soon after closure and extubation, the animals were allowed to resume oral feeding. They were followed for clinical status, weight gain, and endoscopic studies. They were electively sacrificed after six weeks to four months.

Results: All ten animals survived long term, giving success rate of 100% (10/10). One animal developed tension pneumothorax early after surgery, which was properly treated. No other complication occurred. They all resumed oral feeding soon after the surgery, and gained weight. No infection was encountered. Endoscopic studies showed complete mucosal coverage by six weeks. There was minimal to no anatomical or functional stricture at the repair site. Excised specimens after sacrifice showed complete mucosal coverage of the repair site, with regeneration of the all three layers.

Conclusion: Our Elastin based biomaterial patch can be used safely and reliably for repair of esophageal injury with significant tissue loss. Complete regeneration of the esophageal wall was observed when repaired immediately after the injury in our experiment.

PERFORMANCE OF A SCALED-UP PACKED-BED REACTOR FOR USE AS A BIOARTIFICIAL LIVER

[Purpose] A technique for large-scale and high-density culture of hepatocytes is essentially required for the realization of a bioartificial liver. In an earlier study, we reported that the packed-bed reactor using highly porous polyvinyl formal (PVF) resin is a promising means to attain the high-density culture of hepatocytes. In the present study, the reactor was scaled up from 4 cm^3 of prototype to 67.5 cm^3 in culture volume and metabolic performance of the reactor was evaluated.

[Methods] A packed-bed reactor (50 mm i.d.) loaded with 2,500 PVF resin cubes (3 x 3 x 3 mm; mean pore size, 250 μm) was used. Hepatocytes ([2.5 ~ 4.7] x 10^9 cells) obtained from pigs were inoculated into the reactor. Five hundreds milliliter of culture medium were perfused from the reservoir into the reactor through an oxygenator at a perfusion rate of 200 ml/min. Monolayer cultures were simultaneously grown on collagen-coated Petri dishes as controls. Metabolic activities of the reactor were evaluated in terms of ammonium metabolism, urea synthesis and albumin secretion of the cultured hepatocytes.

[Results] Immobilized cell densities in the PVF resin were 24 ± 7.6 x 10^6 on Day 2 (n=6) and 16 ± 3.9 x 10^6 cells/cm^6-PVF (mean ± SD) on Day 5 (n=3). The reactor showed satisfactory and stable metabolic activities of the immobilized hepatocytes with measured levels of ammonium metabolism (Km: 2.5 ± 0.96 x 10^{-10} L/cell/h on Day 2) and albumin secretion (0.71 ± 0.39 μmole/10^6 cells/h) comparable to those obtained in monolayer (Km: 3.6 ± 1.6 x 10^{-10} L/cell/h, albumin: 1.6 ± 0.53 μmole/10^6 cells/h). TEM observation revealed that the immobilized cells showed well-differentiated ultrastructural morphologies similar to those observed *in vivo*.

K. Yanagi, T. Tun, T. Aung, H. Miyoshi, H. Taniguchi, Y. Takada, K. Fukao and N. Ohshima
University of Tsukuba

A NOVEL TISSUE ATTACHMENT SCHEME FOR BIO-MECHANICAL PROSTHESES

Harnessing in situ muscle power for circulatory support requires a reliable means to anchor a biomechanical actuator to both muscle and bone. The objective of this study was to determine the long-term stability of a tissue attachment scheme designed to secure a piston-based hydraulic actuator to the latissimus dorsi (LD) and underlying ribcage. Woven stainless-steel mesh was used as a porous backing plate for fixing the device to the chest wall (using heavy suture tied through the mesh and around the ribs). LD attachment was made by wrapping the humeral tendon in Dacron fabric and pressing it between toothed clamping plates machined into the piston head. A prototype device was implanted beneath the LD of a mongrel dog and the muscle stimulated for 8 weeks (33 Hz, 9 pulses/burst, >2.6x10^6 contractions) following a 7-day recovery period. Piston position was fixed to maximize forces across tissue/device attachment points. Stimulation produced palpable contractions throughout the implant period. At explant, the device was fully encapsulated in its original location with LD tendon firmly attached. LD adhesions to the device were minimal, but were significant around and above the piston head. Thick fibrous ingrowth through the wire mesh formed a secure bond to the chest wall. A thinner, non-adherent fibrous layer encased the main housing and piston head. Biopsies from LD and surrounding tissues appeared normal under histologic examination. This study suggests that the use of a porous backing plate and Dacron-reinforced tendon clamp mechanism may provide an effective means to anchor motor prostheses for muscular actuation.

Dennis R. Trumble, and James A. Magovern. Allegheny-Singer Research Institute, Pittsburgh, PA 15212

MECHANICAL PROPERTIES OF MESH-TYPE ARTIFICIAL TRACHEAS

S Kawaguchi, T Nakamura, Y Shimizu, T Masuda, T Takigawa, Y Liu, H Ueda, T Sekine, K Matsumoto, N Toba, M Yoshitani, Y. Hori
Kyoto University, Japan

Many studies have been made for mesh-type artificial tracheas and their good performance has been reported recently, but there are few studies of the mechanical properties in spite of importance in practical use. In this study, we determined the mechanical properties of typical mesh-type artificial tracheas and compared them with those of canine trachea.

Four types of artificial trachea were manufactured whose basic frame was composed of a fine Marlex® mesh cylinder and a polypropylene spiral stent. Compression test was carried out on these specimens and the canine fresh trachea in radial direction. Tensile strength of the tracheal cartilage was measured on the rectangular specimen (1.0 × 0.7mm) of canine cartilage.

The specimen whose mesh was sealed with collagen sponge showed almost the same behavior in the force-strain curve under compression, suggesting that collagen sealing has little effect on mechanical properties. Young's modulus of the artificial tracheas could be calculated with the mechanical models. Agreement between measured and estimated mechanical properties was good, especially in the low strain region. This fact means that artificial tracheas can be designed in terms of mechanical properties by mainly considering the basic frame structure.

INFLUENCE OF STATIC PRESSURE TO RED BLOOD CELL DAMAGE UNDER SHEAR RATE CONDITONS

Background: The causes of hemolysis due to physical forces are still unclear. In order to investigate the effect of static pressure and shear rate on hemolysis, we performed in vitro experiment.

Materials and methods: An apparatus consisted of a rotational viscometer, a pressure chamber and a heater was used at in vitro test. The rotational viscometer, which applied shear rate (0, 500, 1500s^{-1}) to bovine blood, was set into the pressure chamber. The pressure chamber was connected to the air compressor to adjust the pressure levels inside of the chamber respectively (0,200,400, 600mmHg). The heater was used to maintain the constant temperature at 28□ inside the chamber. The experiments were performed at 6 times for each condition and the duration was 120 minutes.

Results: It was not observed any significant hemolysis under the condition of the pressure levels between 0 and 200 mmHg and all of shear rates. Hemolysis with shear rate of 1500 s^{-1} was more than six times higher than that of 0 and 500 s^{-1} ($P<0.001$) when the pressure level was 600 mmHg. On the other hand, significant differences in hemolysis were measured between 200 and 600 mmHg ($P<0.001$) when shear rate was 1500s^{-1}.

Conclusions: In conclusion, the results shown that the red blood cell damage depended on shear rate level and pressure level. In addition, it was identified that the damage of red blood cell under high shear rate might effect pressure levels.

T.Yasuda, K. Shimokasa*, A. Funakubo**, F. Miyawaki**, Y. Fukui**, T. Higami***, T. Kawamura****.
Tokyo Denki University Frontier R&D, *Polytechnic University, **Tokyo Denki University, ***Hyogo Heart and Brain center at Himeji, ****Hyogo Health Science Center.

IN VITRO THROMBOGENESIS EVALUATION OF A VENTRICULAR ASSIST DEVICE

Assessment of the thrombo-embolic potential of a ventricular assist device (VAD) is most appropriately carried out *in vivo*. However, because of the time, expense, manpower, and special facilities required for *in vivo* testing, there is need for an *in vitro* screening test for comparison of alternative pump designs.

Fresh bovine blood was collected via vascular puncture and used to prime two flow loops incorporating AB-180 Circulatory Support System Pumps (Cardiac Assist Technologies, Pittsburgh, PA). Pumping commenced simultaneously in both loops at an identical pre-selected RPM and flow rate. Pressure difference (outlet minus inlet), pump current and blood temperature (maintained at 37 ± 2°C) were monitored. Initial and final samples were drawn for ACT, hematocrit, and plasma free hemoglobin measurements. The pumps were stopped after 3 hours, disassembled and photographed; surfaces under each thrombus were examined for imperfections related to thrombus formation.

A total of three pairs of pumps were tested to compare two materials used to fabricate the pump impellers. Each test included a control impeller made from material A, which has an extensive history in both animals and humans. The experimental device of each pair included an impeller made from material B, which is being considered to replace material A.

Two of the three tests resulted in greater amounts of thrombus on material A than on material B. The third test resulted in greater thrombus formation on material B, but only in areas containing clearly identifiable surface imperfections.

The test results indicate that material B, when free from surface imperfections, does not cause an increase in thrombus formation over material A *in vitro* and can be considered for further testing.

AR Abrishamchian*, GA Johnson**, TD Will**, AH Goldstein*
*The Ohio State University, Medical Center. **Cardiac Assist Technologies

WIRELESS BIOSENSORS FOR ARTIFICIAL ORGANS

A revolution in biomedical semiconductor technology is in process, based on the development of 3D integrated circuits (IC). Novel mask-less photolithographic technology has made it possible to map IC onto the surface of single crystal silicon microspheres as small as 0.5 mm. Over 90% of the spherical surface of a 1 mm diam. IC has been mapped, providing 3 times the area available for planar IC mapping. Continuous copper circuit windings have been fabricated on the surfaces of these microspheres, making high inductance R-L-C circuits feasible. Thus wireless, two-way RF communication with microcircuitry implanted within the body is permitted. 3D IC fabrication is achieved in a continuous process, avoiding expensive planar chip wafer foundries. Many types of biomedical microsensors are feasible. Shown is a 1.1mm 3D accelerometer, with a floating ball centered electrostatically within a spherical annulus. Ball displacement by acceleration force in any direction (fluid flow, muscle contraction, etc.) is sensed by a 6 pole capacitance bridge: three pole regions are depicted in the figure. An RF microsphere amplifier with built-in spherical coil antenna (not shown) attaches to gold bumps at the bottom of the sensor. Microsensors of this type may improve the monitoring and control of artificial internal organs.

P. Ahrens[1,2], A.Ishikawa[1], N.Takeda[1], S.Ahn[1], and R.Eberhart[2]
1. Ball Semiconductor, Inc., Allen, TX
2. University of Texas Southwestern Medical Center at Dallas

FLUID-DYNAMIC OPTIMIZATION OF THE *PUMPING OXYGENATOR*: CFD SIMULATION AND IN VITRO TEST.

We recently proposed a new device for extracorporeal circulation, (the *Pumping Oxygenator*, PO) functionally integrating pulsatile pumping and oxygenation of blood. Flat silicone membranes are used both as actuator of a pneumatic pump and gas/blood phase separator of an oxygenator. Pressurized oxygen alternately fills the gas side and blood is pumped through one-way valves. Further improvements of the PO project were conducted both by computational means and by hydraulic tests.

3D Computational fluid-dynamic models were developed with Fluent® (Fluent Inc., Lebanon, NH, USA) to optimize the shape of the blood distributor and collector, connecting the membrane module to the rest of the circuit. Four different shapes were studied, and the best solution was determined basing on the minimization of priming volume, stagnation areas, shear stress and unevenness in the flow delivered to the module. Fluid-structure interaction methods were applied in a 2D model of the blood/membrane region, in order to investigate the local mechanisms of pumping and optimize the device's pumping effectiveness without increasing the mechanical load to blood. A critical phenomenon was observed, consisting in the premature membrane collapse close to the outlet during the systolic ejection. The strategies for avoiding such an effects were pointed out both in design and in regulation terms.

Finally, the previously realized PO prototypes were subjected to *in vitro* hydraulic tests involving a mock loop (representing a patient's systemic circulation) and the pulsatile power output of the device was evaluated at different operating conditions (pulse frequency, blood flow rate, driving pressure and length of the arterial pipes). The results of the tests revealed that the most important influence is exerted by the pipes, whose length is liable to minimization, thanks to the device's intrinsic compactness.

M.L. Costantino, G.B. Fiore, G. Guadagni R. Fumero and F.M. Montevecchi, Dept. Bioengineering, Politecnico di Milano, Italy

REMOVAL OF EXCESS CISPLATIN FROM BLOOD USING AN EXTRACORPOREAL DEVICE WITH IMMOBILIZED ADSORBENT

Cisplatin is presently among the most frequently used chemotherapeutic agents for the treatment of malignant tumors, however the side affects that accompany cisplatin infusion restrict the dose that can be safely applied. Toxicity can be reduced by removing cisplatin leaving the tumor before it is distributed in the general circulation. This can be accomplished using a hollow fiber device developed for extracorporeal removal of cisplatin from blood. The devices employed are converted dialysis cartridges with 10,000 hollow fibers, where cisplatin adsorbent is immobilized behind the fiber membrane. As blood flows through the fibers, circulating cisplatin crosses the pores, and is retained by the adsorbent. A broad range of materials, polymeric adsorbent resins in colloidal form, lipophilic and hydrophilic dextrans were tested for their cisplatin binding capacity and kinetics, at clinically relevant concentrations of cisplatin. Charcoal showed the highest adsorbent capacity: 1044 ug cisplatin per g. Devices prepared with charcoal were tested in in vitro experiments for the ability to remove cisplatin from aqueous solutions and from cisplatin spiked serum. When blood obtained from patients during cisplatin infusion was circulated through the device, 80% of cisplatin was removed. In dog experiments, cisplatin (100 mg/m2) infused into the femoral artery was captured by the device inserted into the femoral vein. Thus restriction of the cisplatin to the area to be treated allows for higher local doses and reduces systemic side effects. Clinical studies with localized cisplatin infusions are being planned.(Supported by the J R Oishei Foundation)

CM Ambrus, C Karakousis, S Khan, A Stadler, Buffalo General Hospital and Hemex, Inc., Buffalo, NY.

DEVELOPMENT OF MEMS-BASED INTRAVASCULAR MICRO ACTIVE CATHETER SYSTEM

We have developed the intravascular micro active catheter system for the microsurgery of the human vascular systems, such as cerebrovascular and coronary artery. This system can be also applied to the laparoscopic treatment. This system consists of a micro active bending catheter and some functional catheters. The micro active bending catheter is a tube-like catheter with 3.0 mm in outer-diameter, 2.0 mm in inner-diameter and 1,000 mm in length. The inner hole of the catheter is used for the pathway of the functional catheters. In the proximal tip of the catheter, an actuator with three shape-memory-alloy springs is installed with plastic and brass connection links. This actuator can be bent to any direction with the PWM controlled electric current. The micro drug infusion catheter, conventional suction tube, micro ultrasonic catheter and angioscopic optical fiber can be used as the functional catheter, and we developed the micro drug infusion catheter suitable for the micro active bending catheter. We tested the developed micro active bending catheter and micro drug infusion catheter with the mock circulatory system. The developed micro active bending catheter can be introduced through 90°, 120°, 180° crooked branches to the brachiocephalic artery, common carotid artery, subclavian artery and so on, in the mock circulation system. Also we tested these catheter system during the laparoscopical surgery of the pig. We could easily induce the catheter to the point we want in the pig ovary.

J.K. Chang, S. Chung, S.K. Lee, S.Y. Yang, S.Y. Moon, and D.C. Han, Seoul National University, Seoul, 151-742, Korea

DEVELOPMENT OF POLYMER MICRO CHECK VALVE FOR CEREBROSPINAL FLUID SHUNT SYSTEM

We developed the micro cerebrospinal fluid (CSF) shunt valve with surface and bulk micromachining technology of MEMS. This micro CSF shunt valve was formed with four micro check valves to have a membrane connected to the anchor with the four bridges. The up and down movement of the membrane made the CSF shunt operation, and the valve characteristics such as open pressure was controlled by the thickness and shape of the bridge and the membrane. The membrane, anchor and bridge layer were made of the O_2 RIE (reactive ion etching) patterned Parylene thin film to be about 5~10 microns in thickness on the silicon wafer. The dimension of the rectangular nozzle is 0.2 * 0.2 mm^2 and the membrane 0.45 mm in diameter. The bridge width is designed variously from 0.04 mm to 0.12 mm to control the valve characteristics. To protect the membrane and bridge in the CSF flow, we developed the packaging system for the CSF micro shunt valve with the deep RIE of the silicon wafer. Using this package, we can control the gap size between the membrane and the nozzle, and protect the bridge not to be broken in the flow. The total dimension of the assembled system is 2.5 * 2.5 mm^2 in square, 0.8 mm in height. We could precisely control the burst pressure and flow rate of the valve varing the design parameters, and develop the CSF shunt system using micro valve fabricated with MEMS technology.

J.K. Chang, S. Chung, J. Kim, S.K Kim, and K.C. Wang, Seoul National University, Seoul, 151-742, Korea

DIFFERENTIATED HUMAN PROMYELOTIC LEUKEMIA CELLS AS A POTENTIAL IMMUNMODULATORY TOOL FOR THE TREATMENT OF PATIENTS SUFFERING FROM SEPSIS

Sepsis is a major cause of death in intensive care units in developed countries. New therapy concepts include extracorporeal immune-support, as e. g. extracorporeal spleen perfusion. New extracorporeal therapy concept with differentiated HL-60 cells has been developed in order to remove bacterial and viral substances by phagocytosis and to act via immunmodulatory effects on cytokines (e.g. IL1; TNFα) and their receptors. - The human promyelotic leukemia cell line HL-60 was differentiated into immuncompetent cells (granulocytes, monocytes) by retinoic acid, G-CSF, vitamin D_3 and interferon γ. Changes in the expression profile of surface markers (CD11b, CD14, CD64, HLA-DR) were investigated by FACS-analysis determining the differentiation mode. Quantitative phagocytosis was measured by using chemiluminescence, phagocytosis tests for flow cytometry and fluorescence phagocytosis with a fluorescence-microscope. Differentiated cells show phagocytosis-capacities as high as normal monocytes and granulocytes. Measurements by FACS-analysis show, that 89 % of differentiated cells induced by retinoic acid, vitamin D_3 and GM-CSF are able to perform phagocytosis. In chemiluminescence the emission of RLU (relative light units) in time of phagocytosis with opsonized zymosan particles was up to 23-fold higher than in unstimulated cells. Emission of cytokines (IL 1β, IL 6, IL 8, IL 10, IL 12, IL 18, TNF α) was measured using ELISA tests. Differentiation into monocytes show e. g. a 110-fold higher emission of IL6 in process of phagocytosis after 6 h compared with gran. differentiation, unstimulated cells show no emission of IL6. First animal trials with rats finished succesfully (ASAIO J. 1999, abstr. 130). Currently we are testing this concept in a porcine model.

Differentiated HL-60 cells are in morphology, function and immunological reactions similar to granulocytes or monocytes respectively. These results demonstrate, that it is possible to produce immunecompetent cells in vitro for extracorporeal immune support strategies as sepsis treatment.

Kleinfeldt T., Altrichter J., Freytag J., Mitzner S., Stange J., Schmidt R. University of Rostock, Celltect GmbH

ESTABLISHMENT OF A HIGHLY DIFFERENTIATED IMMORTALIZED HUMAN HEPATOCYTE CELL LINE WITH TIGHT REGULATION

N.Kobayashi, H.Noguchi, T.Fujiwara, N.Tanaka.
First Department of Surgery, Okayama University Medical School, Okayama, Japan.

Purpose: Bioartficial liver (BAL) is a promising therapy for the treatment of acute liver failure. Primary human hepatocytes are ideal in BAL, however, the shortage of human livers for hepatocyte isolation limits the use of this modality. An attractive solution is to use a clonal cell line that can grow in culture and exhibit the characteristics of differentiated hepatocytes.

Methods: Primary human hepatoctyes were immortalized with a plasmid SV3neo containing the genes of simian virus 40 large T antigen and neomycin phosphotransferase by a method of lipofection. As a safeguard in clinical application, herpes simplex virus-thymidine kinase (HSV-tk) gene was retrovirally introduced into the immortal cells. The gene expression of liver-specific functions and in vitro sensitivity to ganciclovir (GCV) were examined. To investigate the in-vivo potential of cells, intrasplenic transplantation (Isp-Tx) was performed in the rats underwent 90% hepatectomy.

Results: One of the resulting clones, NKNT-10, showed a highly differentiated liver functions and stopped cell proliferation in the presence of GCV. Isp-Tx of $50x10^6$ NKNT-10 cells protected the 90% hepatectomized rats from hyperammonemia and prolonged the survival of them. We have established a tightly regulated immortal human hepatocyte line. The cells can offer the advantages of uniformity, sterility, unlimited availability, and freedom from infectious pathogens and may be useful for BAL.

EXPANSION OF PRIMARY HUMAN HEPATOCYTE POPULATIONS BY RETROVIRAL GENE TRANSFER AND ADENOVIRUS-MEDIATED CRE/LOXP SITE-SPECIFIC RECOMBINATION

N.Kobayashi [1], H.Noguchi [1], T.Fujiwara [1], N.Tanaka [1], K.A.Westerman [2], P. Leboulch [2], [1] First Department of Surgery, Okayama University Medical School, Okayama, Japan. [2] Harvard Medical School-Massachusetts Institute of Technology, MA, USA.

Purpose: Temporary support of a damaged liver by a bioartficial liver (BAL) devise is a promising approach for the treatment of acute liver failure. Human primary hepatocytes are ideal as a source of hepatic functions in BAL, however, the shortage of human livers available for hepatocyte isolation severely limits the use of this modality. To resolve this issue, we have focused on a reversible immortalization system which allows temporary expansion of primary hepatocyte populations by transfer of oncogenes that can be subsequently excised by Cre/loxP site-specific recombination.

Methods: Adult human hepatocytes were immortalized with a retrovector containing the genes encoding Simian virus 40 T antigen (SV40Tag) flanked by a pair of loxP recombination target that was subsequently excised by Cre recombinase. One of the resulting hepatocytes, NKNT-3, showed gene expression of bilirubin UDP-glucuronosyltransferase, glutathione-S-transferase π, blood coagulation factor X, and glutamine synthetase. An adenoviral vector producing Cre recombinase (Ad-Cre) was applied for Cre/loxP recombination in NKNT-3 cells.

Results: RT-PCR and Western blotting studies demonstrated complete elimination of SV40Tag in NKNT-3 cells after Ad-Cre infection. After removal of SV40Tag, morphologically NKNT-3 cells had similar characteristics of primary hepatocytes and gene expression of liver-specific functions was enhanced. We have established a reversible immortalization system in primary adult human hepatocytes. This work represents a novel strategy for resolving the organ shortage that limits the use of human primary hepatocytes for BAL treatment.

DEVELOPMENT OF AN IMPLANTABLE ARTIFICIAL ANAL SPHINCTER BY THE USE OF THE SHAPE MEMORY ALLOY

After the surgery after the rectal cancer, almost patients must have the stoma. It was not so desirable for some patients and loss of quality of life (QOL). In order to improve the situation, we developed an implantable artificial anal sphincter by the use of the shape memory alloy (SMA). By the use of the SMA, we can control an Evacuation. By the use of an animal experiments with the adult goats weighing 60-70 kg, which was the average weigh of the Japanese, an artificial anal sphincter with SMA was evaluated. Average internal pressure of the rectum and colon were thought to be almost 50 mmHg or so, which was evaluated in the clinical measurement. So, artificial feces with 50 mmHg pressure was tested in the animal experiment. As the results, satisfactory basic performance was recognized in the experiments. We have been developing the transcutaneous energy transmission system (TETS) for an implantable artificial heart. If we combined an implantable artificial sphincter and TETS, totally implantable artificial anal sphincter will be embodied. This system is now under developing. Following evaluation aiming at clinical application may be continued in future.

T.Yambe, S.Amae, H.Takagi, T.Kamiyama, Y.Luo, S.Nitta, Tohoku University

WEAR EVALUATION OF CARBON AND NITROGEN TREATMENTS OF TI-6AL-4V IN THE VENTRASSIST ROTARY BLOOD PUMP

The VentrAssist rotary blood pump is an implantable centrifugal pump with a hydrodynamically suspended rotor/impeller. The pump housing and impeller casing are constructed from Ti-6Al-4V; treatment of the pump surfaces was examined to enhance biocompatibility and reduce friction at pump start-up.

One carbon surface was applied by sputter deposition (CS). Three alternative treatments applied by plasma immersion ion implantation were examined: one was carbon (CP), one was nitrogen (NP) and one was titanium nitride (TiNP). Treatments were compared with control untreated Ti-6Al-4V using pin on disk wear testing, with 1 N load at 2,000 rpm.

The coefficients of friction for the different sample disks, run against control pins, were: 0.07, 0.08, 0.11, 0.18 & 0.18, for the CS, CP, NP, TiNP and control samples, respectively. The coefficient of friction for the CS pin against the CS disk was better than that against the control disk, whereas the coefficient of friction for all other matched surfaces was equivalent, or worse, than that obtained when tested against control surface. Carbon coating reduces friction and is expected to enhance biocompatibility.

N.L.James, J Kaur, R.Tarrant*, D.McKenzie*, M.V. Swain+, G.Tansley, J. Woodard
VentrAssist, Sydney, Australia.
*Applied Physics, University of Sydney; +Biomaterials Research Unit, University of Sydney, Australia.

FLOW VISUALIZATION TO EVALUATE HEMOLYSIS IN CENTRIFUGAL BLOOD PUMPS

1. Introduction: We developed an improved visualization method to observe continuously the relative flow within centrifugal impellers driven at high rotational speed. Applying the method in this study, we visualized the flow of pump models with different impeller configuration to find out the relationship between flow pattern and hemolysis.
2. Methods: Three pump models with different vane configuration were evaluated. An optical based tracer method utilizing a high-speed video camera and a device that keeps the image of impeller rotating at several hundred rpm stationary was applied to visualize flow patterns. In this method, movement of a specific tracer particle in the fluid could be monitored continuously for a sufficient duration and the locus of the tracer particle representing the flow pattern relative to the impeller was reconstructed by using computer assisted image measurement. Configurations of the three types of investigated impeller were Type A; six straight vanes had an inlet angel of 57.5 degrees and an outlet angle of 82.5 degrees. Type G, six curve vanes inlet angel of 60.0 degrees and an outlet angle of 120.0 degrees; Type I six curve vane both the inlet and an outlet angle of 60.0 degrees. Particles made of activated spherical charcoal were used as tracer particles for flow visualization.
3. Results: The motion of tracer particles from the inlet portion to the outlet portion of the impeller was analyzed. Even if the pumps were operated at almost the same rotational speed under the same flow rate and pressure head, differences in the flow pattern with the differences in impeller configuration were observed. These showed that the particles undergo an irregularly fluctuating stress loading and the stress loading has peculiarities in each type of impeller. This peculiarity is considered to have close relationship with mechanical blood damage that deteriorates pumps hemolytic performance.

S. Ahmed, A. Funakubo, I. Sakuma**, F. Miyawaki*, Y. Fukui*, T. Dohi***
Tokyo Denki University, Frontier Research and Development Center.
** Tokyo Denki University. ** Graduate School of the University of Tokyo.*

DEVELOPMENT OF A BALLOON DRIVE SYSTEM FOR THE INTRAVASCULAR MEMBRANE OXYGENATOR

Gas exchange for the University of Pittsburgh Intravascular Membrane Oxygenator (IMO) depends on the convective blood flow generated by the pulsating balloon at the center of the device. Recent design changes in the IMO include the use of a smaller balloon which must be pulsated at higher frequencies to generate the same convective blood flow as the original 40cc balloon. A new drive system is being developed to pulse balloons with volumes from 10 to 40cc at frequencies from 600 to 60 beats per minute. Two approaches are being investigated simultaneously: a pneumatic drive system that includes a compressor and two solenoid valves, and a direct drive system comprised of a servo motor, actuator, and bellows. The intra-aortic balloon pump originally used for balloon pulsation could drive a 40cc balloon at 150 beats per minute, and a 25cc balloon at 180 beats per minute. A preliminary version of the pneumatic balloon drive system pulsates a 40cc balloon at 180 beats per minute, and a 25cc balloon at 240 beats per minute. The pneumatic and direct drive systems are evaluated based on their ability to generate convective flow irrespective of balloon volume. The strengths and weakness of the two systems are compared and contrasted to determine which is more appropriate for the IMO project.

T. Tulou, B.G. Hattler, W. Federspiel
University of Pittsburgh

KINETICS OF ANTIBODY REMOVAL BY DIALYSIS MEMBRANE IMMUNOADSORPTION

Therapeutic antibody removal can be used to alleviate the symptoms of autoimmune diseases and to prevent humoral rejection following organ transplant. Using conventional methods of antibody removal necessary plasma components are removed in addition to the targeted antibodies. Membrane immunoadsorption represents a potential antibody removal therapy in which only the targeted antibodies are removed. We have previously shown the feasibility of this technique by removing anti-A and anti-B antibodies from whole blood using hemodialyzers with antigen attached to the hollow fiber membranes. In this study we analytically and experimentally determined the dependence of the antibody removal rate on the membrane geometry and blood flow conditions.

We developed an analytical model to predict the rate of antibody removal when the concentration of free antibodies at the fiber wall is negligible. The model predicts that the antibody concentration drops over time according to a time constant which depends on the total blood volume, the antibody diffusion coefficient, the blood flow rate, and the filter geometry. The predicted time constant for the anti-A and anti-B capture experiments (24 min) was of the same order as the measured time constant (15 min). To obtain more precise data we developed a model immunoadsorption system in which hemodialyzers modified by attachment of triazine dye are used to remove bovine albumin from solution. Data from these capture experiments provides additional validation of the analytical model.

M.S. Hout and W.J. Federspiel
University of Pittsburgh McGowan Center for Artificial Organ Development

ARTIFICIAL LUNG PROTOTYPE DEVELOPMENT

S. D. Lick[1], S. K. Alpard[1], P. Montoya[2], D. J. Deyo[1], J. B. Jayroe[1], J. B. Zwischenberger[1]
University of Texas Medical Branch, Galveston, Texas[1], University of Michigan Medical Center, Ann Arbor, Michigan[2]

We are developing a low-resistance artificial lung for ambulatory medium-term use (weeks to months) as a potential bridge to transplant or recovery using a sheep survival model.

Ventricular assist device (VAD) arterial cannulae are anastomosed to the proximal and distal main pulmonary artery, and an inflatable occluder is placed between, which, when inflated, diverts full main pulmonary artery flow through the artificial lung. The cannulae attach via 5/8" connectors to a rigidly housed, low resistance, microporous hollow fiber membrane gas exchanger (artificial lung, MC3, Inc. Ann Arbor, MI: 2.25m^2, priming volume 350cc).

8/8 animals have survived the implant. Two were electively sacrificed after 7 days. The others died either of late exsanguination (3 sheep), insanguination into the chest from a late anastomotic dehiscence (surgical technique error,1 sheep), or right heart failure (two sheep). Causes of exsanguination were broken blood ports (2 deaths) or cannula disconnection (1 death). Based on this experience, we have made 3 separate modifications: 1) removal of blood sampling ports; 2) change from slip-on cannula connectors to a more secure collet-nut configuration; and 3) lower resistance artificial lung (mean gradient at total PA flow from 8 mmHg to 6 mmHg) by improving blood flow patterns.

A rigidly housed, low-resistance gas exchanger can function as a chronic artificial lung when interposed to accomodate total pulmonary artery blood flow.

PORCINE MODELS OF EXPERIMENTAL FULMINANT HEPATIC FAILURE

A glass bead packed-bed bioreactor (**BR**) has been developed for extracorporeal liver assist. The system is readily scalable, but the requisite hepatocyte mass for adequate liver function for recovery or bridge to transplant is unknown. To determine this number, and to critically evaluate the **BR**, a reversible large animal model is needed that reproduces the pathophysiology of fulminant hepatic failure (**FHF**). The model should mimic necrotic liver with early regeneration, and maintain normal fluid and renal physiology. Surgical/ischemic models do not meet these criteria. We evaluated two models of chemically induced **FHF** in pigs: 1) acetaminophen (**Ac**), and 2) d D-galactosamine (**dGal**).

Method: Model 1) Five animals were pretreated with buthionine sulfoximine before dosing them with **Ac** over a 12 hr period. Model 2) Six animals received a single dose of **dGal**. The animals were followed clinically and biochemically with serial prothrombin time (**PT**), serum ammonia (**NH3**) and transaminase levels (**TA**) for evidence of liver failure.

Results: The animals from both models quickly developed signs of **FHF**, but the pigs given **dGal** followed a more predictable pattern. In the **Ac** model five animals were consistently lethargic at 6-8 hr, vomiting followed, and by 12 hr they were somnolent. Three animals were sacrificed at 18, 20, 24 hr respectively. Two others showed less profound clinical changes with signs of recovery beginning at 48-60 hr. Lab results reflected liver damage but not the clinical severity. **TA**, especially LDH and AST, were raised in all pigs. Prolonged **PT** was seen but was profound in only one. Histologically, the liver showed extensive centrilobular injury with relative sparing of the periportal area of the lobule.

In the **dGal** model there is an early latent period of 48 hr. Neurological signs of **FHF** developed slowly. **NH3** was consistently elevated when neurological changes were well established. Coagulopathy, starting at 36 hr, persisted, and **TA** increased until death. Bilirubin was elevated. Histology was typical of acute liver damage with panlobular hepatocyte ballooning and destruction being uniform throughout.

Summary: Pigs given a single dose of **dGal** form an excellent model of FHF and is useful to study and evaluate various treatment options.

V. Ramachandran, T.D. Whitehead, A. Ulivargoli, J.A. Lowell, S. Shenoy, T. Mohanakumar, T.K. Howard
Washington University School of Medicine, St. Louis, MO

3-D COMPUTATIONAL FLUID DYNAMICS IN ARTIFICAL LUNG DESIGN: STEADY AND TRANSIENT MODELS

A 3-D Computational Fluid Dynamic (CFD) model of a previously described total artificial lung (TAL) was developed. The model was developed to optimize the TAL design based on three parameters: flow distribution, pressure drop (ΔP), and lung impedance in pulsatile flow. METHODS: ADINA (ADINA R&D, Inc., Watertown, MA), a finite-element based package, was used in this work. The fiber bundle was assumed to be a porous medium, and thus, modeled using Darcy's equations. The porosity (ϕ=0.5) was calculated from the fiber geometry and spacing, and the permeability ($k=7.4e^{-9}$ m^2) was measured experimentally. For this preliminary work, the fluid was water (ρ=1000 kg/m^3, μ=0.001 Ns/m^2). For steady-state flow, ΔP was prescribed as the boundary condition. For transient flow, physiologic inlet and outlet pressures were prescribed as functions of time. RESULTS: For the initial TAL design, the calculated steady-state flow rate (Q) vs. ΔP was within 5-20% of the measured Q up to 10 l/min. Several design alterations were studied, and the CFD results trended with the measured Q vs. ΔP. For the transient model, the calculated Q(t) was within 10-15% of that measured. CONCLUSIONS: These preliminary results suggest that CFD will be a useful tool in hemodynamic optimization of the TAL. Also, the transient model will allow for the TAL impedance to be optimized, while minimizing the need for trial and error device prototyping.

SD Chambers, RH Bartlett, JP Montoya
Michigan Critical Care Consultants, Inc., Ann Arbor, MI

ATTENUATION OF HEMOGLOBIN (Hb) NEUROTOXIC POTENTIAL BY CROSS-LINKING WITH ADENOSINE AND CONJUGATION WITH REDUCED GLUTATHIONE (GSH)

Chemically modified Hb solutions are under development as red cell substitutes. During a recent clinical trial, however, one of these products was found to be involved in serious adverse events in patients with acute ischemic stroke (Stroke 30:993-6;1999). Therefore, it has been recommended that the safety evaluation of any new blood substitute candidate should also include the study of its neurotoxicity. As a result, we have investigated the neurotoxic potential of our developed blood substitute, consisting of bovine Hb cross-linked intramolecularly with *o*-ATP and intermolecularly with *o*-adenosine and conjugated with GSH, on primary cultures of normal human brain neurons and astrocytes. The effect of this blood substitute used in concentrations of 0.1, 1, 10 and 100 μM (as a tetramer) was compared with the effects seen with the native Hb. In the control group, Hb was replaced with plasma. In addition, cells were incubated with both Hb solutions, supplemented with a 10 molar excess of hydrogen peroxide (H_2O_2). After 18 hours of incubation, the cells were tested for viability (LDH release) and oxidative injury (formation of lipid radicals, lipid hydroperoxides, thiobarbituric acid reactants, and the level of intracellular GSH). The astrocytes were also screened for intracellular adhesion molecule-1 expression. Exposure to native Hb, resulted in a concentration-dependent neuronal death and the oxidative and inflammatory activation of the astrocytes. The toxicity of this Hb toward human neurons was potentiated by the presence of H_2O_2. Neuronal death occurred when the concentration of the native Hb exceeded 10 μM, and at 1 μM with the addition of H_2O_2. The astrocytes showed to be more resistant than the neurons. The oxidative injury and inflammatory response were observed at a 100 μM level when the native Hb was oxidatively activated by H_2O_2. On the contrary, our blood substitute showed to be non toxic to the neurons and astrocytes at any tested concentration, and it significantly reduced the toxic effect of H_2O_2. Together, these results indicate that the native Hb is extremely toxic to the neurons, and this effect is potentiated by H_2O_2. This data also demonstrates that the proper chemical modification procedure, which combines Hb molecules with well known neuroprotective agents, adenosine and GSH, effectively attenuates this deadly potential.

J. Simoni, G. Simoni, D.E. Wesson, P. Malek-Ahmadi, J.A. Griswold & M. Feola, Texas Tech University Health Sciences Center, School of Medicine

A RHEOLOGICAL IN VITRO DEVICE FOR INVESTIGATING THE HEMOCOMPATIBILITY OF BIOMATERIALS: RHEOCOAGULOMETER

K. Mottaghy, S. Kirberich, H. Imhoff and A. Henseler
Dept. Physiology, IZKF "Biomat.", University Hospital RWTH Aachen, Germany

The hemocompatibility of artificial surfaces is in tight connection to the fluid dynamic characteristics of flowing blood. In addition to surface effects the growth rate of thrombi, cellular activation and adsorption of proteins are affected by blood flow conditions. We introduce an in vitro flow model („Rheocoagulometer") which allows coagulatory and cellular investigations with defined blood flow properties exposed to biomaterials based on Couette-flow principle. Two exchange-able co-axial cylinders rotate independently producing sheared laminar blood flow in the gap of both with and without secondary flows (Taylor-vortices). On example of polystyrene cylinders could be shown that the plasma protein adsorption was pronounced using laminar shear flow (G=600 s^{-1}). Platelet drop as well as ß-Thromboglobulin/PF4 release was surprisingly less affected in the presence of vortices in comparison to pure shear flow at the same shear rate. On the other hand flow cytometrical results demonstrate a higher activation of the fibrinogen receptors in the presence of secondary flows. The results demonstrate that the device opens a wide range of applications to investigate the complex blood/material interaction under specific rheological conditions for any given biomaterial and possible surface modifications.

HEMOCOMPATIBILITY OF TUBULAR POLYMERS INVESTIGATED WITH A MODIFIED CHANDLER LOOP

K. Mottaghy, B.Oedekoven, C. Andersson and A. Henseler
Dept. of Physiology, IZKF "Biomat.", University Hospital RWTH Aachen, Germany

The choice of an appropriate in vitro method for hemocompatibilty studies of extracorporeally used tubes is still a actual problem. Static procedures may have some advantages such as easy handling but ignore the real situation for blood flow. Therefore a modified Multi Channel Chandler Loop System (MCCL) was designed allowing hemodynamically standardized investigations. Aim of the present study was to determine influences of relevant experimental boundary conditions to be considered such as anticoagulation, age and storage of blood, surface area/volume ratio, shear rates and blood/material contact times.

Static investigations ($G = 0\ s^{-1}$) were compared to those applying clinically relevant shear rates calculated for extracorporeal tubing ($G=30$ to $390\ s^{-1}$) using medical grade silicone and PVC as typical tube polymers. Relevant cellular and plasmatic parameters such as PF4, TAT, C5a were measured.

The results indicate:

- In contrast to the dynamic measurements, at shear rate $G = 0\ s^{-1}$ (static condition) no significant differences for various parameters were obtained.
- Even one hour storage of freshly drawn donor blood in PVC blood bags impaired the validity of the obtained data in contrast to freshly drawn syringe blood.
- Silicone tubing activated platelets and complement in a higher degree than PVC tubing.
- Increase of shear rate increased complement activation.
- Shear dependent platelet activation could only be observed by applying blood of the same individual donor.

THE USE OF CELLULOSE BINDING DOMAIN-PROTEIN A FOR IMMUNOADSORPTION OF IgG

Sujatha Karoor[1], Mari Katada[1], Jeanette Molina[1], Jason Wheeler[1], Clark K. Colton[2], and Lee Henderson[1]
[1]*Baxter Healthcare Corporation, McGaw Park, IL*
[2]*Dept of Chemical Engineering, Massachusetts Institute of Technology, Cambridge, MA*

Immunoaffinity adsorption is increasingly used for protein purification and extracorporeal therapeutic applications. Immunoadsorption of IgG is under study for treatment of patients with high panel reactivity prior to transplantation. Synthetic membranes have advantages as support matrices compared to conventional gel media because they are incompressible, eliminate internal diffusion limitations, and permit plasma separation and immunoadsorption in the same device. The goal of this study was to explore the performance of the conjugate cellulose binding domain (CBD)-protein A bound non covalently to cellulose triacetate membranes for adsorption of IgG and to compare it with protein A attached covalently to polysulfone and nylon membranes. The binding constant for CBD-protein A interaction with IgG was measured with both static and dynamic methods. Breakthrough curves up to ligand saturation were measured to study the effect of IgG concentration, filtrate flow rate, and presence of albumin. Experiments with whole blood showed in-vitro feasibility of the concept. Binding capacity increased from 50 to the extraordinarily high value of 400 mg/ml membrane volume with CBD-protein A bound to cellulose triacetate fibers. The capacity was greater by a factor of 15 when compared to protein A attached to polysulfone and nylon membranes.

CFD STUDIES OF C1E3 GYRO CENTRIFUGAL BLOOD PUMP

S. Schulte-Eistrup, T. Takano, T. Maeda, K. Nonaka, J. Linneweber, S. Kawahito, T. Sato, A. Fujisawa*, K. Minami‡, R.Koerfer‡, Y. Nosé
Baylor College of Medicine, Department of Surgery, Houston, TX
*Kyocera Corporation, Kyoto, Japan,
‡Herzzentrum NRW, Bad Oeynhausen, Germany

A double pivot bearing Gyro pump C1E3 has been developed for cardiopulmonary bypass purposes as well as intermediate and long-term centrifugal ventricular assistance.

In long-term ex-vivo animal studies, with an ACT around 200-250 sec, firmly attached fibrin deposit was detected at the bottom shaft after two days of pumping; whereas, no deposit was found at the top shaft of the impeller. However, no signs of thrombembolism were detected in the calves, or clinically, nor macroscopically or histologically.

In order to characterize blood flow in the C1E3 pump computational fluid dynamic studies (CFD) were undertaken. CFD proved that there is effective wash out at the top shaft of the impeller as well as the top female bearing. Even though the fluid velocity, which increases to the periphery, was rather low. The CFD further demonstrated secondary blood flow at the bottom side of the impeller generated by secondary vanes leads to an effective blood clearance. Blood velocity was centrally the lowest, increasing peripherally.

In vitro testing revealed excellent anti-blood-traumatic features with a Normalized Index of Hemolysis (NIH) of 0.0007 g/100L in an LVAD condition. Ex-vivo studies confirmed these data with average free plasma hemoglobin in 5 calves of 4.15 mg/dl +/- 2.7. The CFD studies showed very low shear rates at the impeller corpus with just small areas of high shear stress in the peripheral portion of the top impeller vane. High shear stress was demonstrated at the introduction of the outlet port, where the blood velocity is the highest.

Conclusion: CFD studies confirm the high anti-traumatic features with low shear rates in the housing and impeller. Furthermore, blood wash out from the impeller bottom with low flow centrally as well as at the top and bottom male bearing occurs. These areas have to be further investigated to eliminate any deposition of fibrin.

LIVER REPAIR AND HEMORRHAGE CONTROL USING LASER WELDING OF LIQUID ALBUMIN IN A PORCINE MODEL

Yasmin Wadia, MD, Hua Xie, MD, Michio Kajitani, MD, Scott Prahl, Ph.D., Kenton Gregory, MD
Oregon Medical Laser Center, Providence St. Vincent Medical center, Portland, Oregon 97225

PURPOSE OF STUDY: To evaluate laser soldering using liquid albumin–ICG solder for welding liver lacerations and sealing raw surfaces created by segmental resection of a lobe. Major liver trauma has a high mortality due to immediate exsanguination and a delayed morbidity and mortality from septicemia, peritonitis, biliary fistulae and delayed secondary hemorrhage.

METHODS: Fourteen laceration injuries (6cm long x 2cm deep) and 13 non-anatomical resection injuries (raw surface 6cm x 2cm) were repaired. An 805nm laser was used to weld 53% liquid albumin-ICG solder to the liver surface, reinforcing with a free autologous omental scaffold. The animals were heparinized to simulate coagulation failure and hepatic inflow occlusion was used for vascular control. For both laceration an resection injuries, sixteen soldering repairs were evaluated at three hours, two at two weeks and nine at four weeks. A single suture repair of each type was evaluated at three hours.

RESULTS: All 27 liver repairs were accompanied by minima blood loss as compared to the suture controls. No dehiscence o any laser repair was seen. All 11 chronic repairs healed withou evidence of biliary fistula or secondary hemorrhage.

In conclusion laser welding repair of the liver is a quick and reliable technique for repairing major hepatic trauma.

ALTRUISM, VOLUNTEERING FOR ARTIFICIAL VISION EXPERIMENTATION, AND UNANTICIPATED PSYCHOLOGICAL BENEFIT

Beth J. Seelig and William Dobelle

A) PURPOSE OF STUDY

Psychiatric assessment of volunteers for potentially hazardous experimentation includes psychodynamic assessment of the motivations underlying the altruistic action of volunteering. The goal is to select experimental subjects who will not be likely to be psychologically harmed by participation. This is in addition to evaluation of adequacy of informed consent as well as screening out substance-abusing, psychotic, and other severely mentally ill individuals. We are presenting data from a small group of original volunteers for the "Dobelle eye" artificial vision project who entered the project over twenty years ago, at a time when they could expect no tangible benefit from participation.

B) METHODS USED

Semi-structured psychodynamic interviews of the original group of blind volunteers were conducted, recorded, and transcribed. The original interview data are reassessed from a psychodynamic perspective. Superficially altruistic behavior, such as volunteering for this experimentation, serves multiple psychological functions and in a given clinical case, the determinants are frequently complex. Psychodynamic exploration necessary to sort out the various components is presented. A spectrum of altruistic behavior is suggested, based on data from these original subjects as well as from patients interviewed in the setting of psychoanalytic treatment.

C) RESULTS

In light of our work with the volunteers for this project as well as with other altruistic individuals, we reexamine and propose an expansion of the current understanding of altruism. We suggest that adaptive altruism can explain the unexpected positive finding that some volunteers gained actual psychological benefit from their participation. This finding, that participating in research as an experimental subject can result in lasting improvement in self-esteem, is discussed. Suggestions are made for increasing the likelihood of such benefit. Ethical ramifications are addressed.

SUSTAINED SKELETAL MUSCLE POWER FOR CARDIAC ASSIST DEVICES: IMPLICATIONS OF METABOLIC CONSTRAINTS.

A device to harness power from skeletal muscle contracting in a linear configuration is under development. This application requires a sustained level of power, that is dependent on muscle mechanics and metabolic properties. A biomechanical muscle model and a metabolic model constructed from experimental data were used to predict maximum power available in a sustainable region of loading and stimulation conditions.

Latissimus dorsi (LD) of four goats were evaluated *in vivo* following a 10 week *in situ* conditioning protocol with an implanted Telectronics myostimulator. The LD insertion was reconnected to a hydraulic loading system allowing isometric and isotonic contractions for biomechanical characterization. Metabolic utilization was measured by a thermister based myothermic technique. Brief fatigue tests of working isotonic contractions revealed stimulation conditions associated with sustained power.

The results show metabolic utilization was dependent on contraction duration, rate, force and work. The region of sustainable contractions was found for a range of durations of 0.1 to 0.5sec and rates of 10 to 120 bpm. The boundary for the sustainable power region was well approximated by a constant value of metabolic utilization. A constant duty cycle (contraction to cycle duration ratio) also showed a similar trend but differed by as much a 30%.

The results demonstrate that maximum sustained power can be predicted by a mechanical muscle model when the operating conditions are constrained to a sustainable range determined by a metabolic model. In conclusion, metabolic constraints influence the optimum conditions for sustained power needed in the design of skeletal muscle powered assist devices.

SH Reichenbach[*†], GD Egrie[*], SM Marinache[*], KJ Gustafson[*], DJ Farrar[†], JD Hill[*].
California Pacific Medical Center[*], Thoratec Laboratories[†]

EFFECTIVE REDUCTION OF BLOOD CELL DAMAGE BY USING A COMPLIANCE CHAMBER IN A PULSATILE EXTRACORPOREAL LIFE SUPPORT SYSTEM

Kyu Back Lee*, Kyung Sun*, Hye Won Kim*, Jun Keun Chang**, Chan Young Park**, Chahil Chung**, Jung Kyung Kim**, Jongwon Kim***, Hyoung Mook Kim*
*Korea University College of Medical, **Seoul National University College of Engineering, ***Biomedlab Co. Seoul, Korea

So far many extracoporeal life support systems (ECLS) were shown in the market, but types of their main pumps were either rotary pumps or centrifugal pumps. Pulsatile pumps were usually not used due to the high hemolysis. However, pulsatile pumps have a major advantage that they are capable of synchronizing with the natural heart beat that helps to increase the coronary blood flow and decrease the load of natural heart. In this study, we tried to improve the hemolysis problem of pulsatile ECLS with a compliance chamber located between the outlet of blood pump and the inlet of membrane oxygenator. The role of the compliance chamber was simulated with a mathematical modeling of human vascular system and ECLS system. A set of animal experiment was also performed. Experimental animals were divided into 3 groups. A pulsatile pump with a compliance chamber, a pulsatile pump without a compliance chamber and a centrifugal pump without a compliance chamber were used in group 1 (6 dogs), group 2 (6 dogs) and group 3 (4 dogs) respectively. We observed the pumping capacity, gas exchange capacity and the degree of hemolysis in the acute pulmonary insufficiency animal model. After the onset of acute pulmonary insufficiency by the slow infusion of oleic acid, we started partial cardiopulmonary bypass(pCPB) and finished after 2 hrs. of pCPB. The concentration of plasma hemoglobin was measured before the animal experiments and after 2 hrs. of pCPB. We also observed several hemodynamic parameters and gas exchange parameters. The results of mathematical modeling showed us that the more the amount of air in the compliance chamber, the less the fluctuation of the AoP. The results of animal experiments show us that the hemodynamic parameters of pCPB, such as arterial pressure, central venous pressure, left atrial pressure and pulmonary capillary wedge pressure, were not significantly different among experimental groups. However, the parameters correspondent to the efficiency of gas exchange and the blood cell damage were significantly different among groups. After 2 hrs. of pCPB, the oxygen partial pressures of arterial and mixed venous blood significantly increased in group 2 compared with those in group 3. The concentration of plasma free hemoglobin also increased significantly in group 2 compared with those in group 1 and group 3 ($p<0.05$). The results of mathematical modeling also showed us that the more the amount of air in the compliance chamber, the less the fluctuation of the AoP. In the conduit between the blood pump and the membrane oxygenator, there was an extremely large fluctuation of pressure when the compliance chamber was not used. The fluctuation of pressure could be adjusted to a tolerable level by using a compliance chamber that damped the highest pressure and the lowest pressure. As a result, it decreased the extent of blood cell damage markedly to the level of using centrifugal pump. Furthermore, it lowered the work load of the ECLS device.

AN NEW APPROACH ON RADIAL EXPANSILE FORCE MEASUREMENT OF SELF-EXPANDABLE ESOPHAGEAL METAL STENTS

Taegyun Moon*, Hoon Jai Chun**, Yoon Tai Jeen**, Jin Hai Hyun**, Daehie Hong*, Kyu Back Lee***
*Dept. of Mechanical Engineering, Korea University College of Engineering, **Dept. of Internal Medicine, Korea University College of Medicine, ***Dept. of Biomedical Engineering, Korea University College of Medicine, Seoul, Korea

The expansile force is a marginally important property for the selection and the design of a stent. There have been a few studies on the measurements of esophageal stent's expansile force. The physical properties that they measured, however, were not applicable to the clinical situation in the aspect that they measured neither the true radial force nor the expansile pressure, although the intraluminal pressure and the compliance of a patient's esophagus can be measured with currently available esophageal manometry. We developed a novel method of measurement that closely simulates the real implanting environment. We could measure the true radial force as well as the pressure with this method.

A pair of square blocks with a hemi-cylindrical trough was used to compress a stent uniformly in radial direction. The stent was installed between two blocks and the expansile force was quasi-statically measured with universal tensile testing machine when the blocks started to part. In order to get the relations between the expansile force and the radial deformation, a series of differently sized cylindrical block pairs were used for each stent. Six kinds of esophageal stents; covered and uncovered Y2P SR, covered Y3E SR (Stentech, Korea), covered and uncovered Ultraflex(Boston Scientific, USA) and uncovered NiTi (Tae Woong, Korea), were measured in this study.

The expansile forces measured with this system were observed to be somewhat bigger than those with the conventional methods[1]. This is because the new method is more realistic to implanting situation than the previous simplified methods. Since the actual radii of deformed stents were given in the test, the true pressure was easily calculated with the expansile force. In general, the expansile force was gradually increased as the radial deformation was increased. However, covered Y2P SR and covered Y3E SR showed a sudden drop of force in the middle of the force-deformation curve. We think that the weaving pattern and the coverage of those particular stents caused the unusual behavior. This phenomenon is very important due to the fact that the stent could be locked at this radius before it is fully expanded. Also, the expansile force-deformation curve showed nonlinear behavior when the deformation was large. For this reason, the actual expansile force for the radii in the clinical situation rather than the calculated force from the hoop strength should be measured. We expect that the data of the new method are directly applicable to the clinical selection of stents used in esophagus, in biliary tracts, and in various hollow organs where obstructive illnesses occur. We also expect that they can be effectively used for the stent design.

REFERENCE

1. Chan, A. C. W. et al, Gastrointest. Endosc., 49:462-465, 1999

PLATELET/SURFACE INTERACTIONS: AN *IN VITRO* METHOD FOR SCREENING NEW NONTHROMBOGENIC SURFACES

The development of nonthrombogenic artificial surfaces for extracorporeal circulation (ECC) is ongoing. *Ex vivo* animal studies are labor intensive and expensive, thus the development of a reliable *in vivo* screening method is essential. The following model utilizing test tubes incubated with ^{111}In labeled platelets demonstrates such a method.

The following surfaces were chosen because of their known platelet surface interactions: glass & PVC (activate platelets), polypropelene (PP) (nonactivator). All studies were carried out using rabbit platelets. The harvested platelets were labeled using a modification to a technique first described by Thakur in 1976. For this model the harvested platelets were incubated with 30 μCi ^{111}In : 2.0x10^6/mm^3 platelets for 10 minutes to obtain a 60-80% labeling efficiency. The plasma harvested was retained; the Ca^{2+} was readjusted and the radiolabeled platelets were then resuspended in the plasma to produce a count of 1.6x10^5/mm^3. Two mL aliquots of the above were placed in each tube and incubated for 2 hr at 37^{0}C. Free indium levels, platelet counts, and aggregometry were performed pre and post incubation. Post incubation, test tubes were gently rinsed and placed in a gamma counter to determine platelet adherence. Results of PVC and PP were compared against glass. Validation of the method was demonstrated via platelet metabolic inhibition studies and plasma free indium testing.

For all studies performed, platelets demonstrated functionality to collagen, γ thrombin, and ADP pre and post labeling and incubation, except for the metabolic inhibition studies where there was no response to any agonist as expected. Platelet counts and pre and post free indium levels were similar within each test tube across all 3 groups. Platelet adhesion was greatest on glass (n=12) with PVC (n=16) at 67±8% and PP (n=12) at 43±5% when compared to glass. Metabolic inhibition reduced active platelet adhesion significantly in the glass and PVC groups ($p<0.0001$) but did not change the passive adherence in the PP group ($p=0.19$). Free indium within the plasma did not significantly affect resultant gamma counts ($p>0.1$).

With these results, the model can now be used as a screening test to select those nonthrombogenic surfaces that warrant further *ex vivo* animal and clinical testing.

GM Annich, B Ashton, SI Merz, DO Brant, RH Bartlett
University of Michigan and MC3 Inc.

EFFECT OF SHEAR STRESS ON PLATELET ADHESIONS TO MATERIALS WITH VARIOUS WETTABILITY

To evaluate anti-thrombogenic materials, static systems without shea stress have been frequently used. However, thrombus formation in viv proceeds under blood flow conditions. Therefore, we investigated whethe shear stress was essential for evaluating *in vitro* the hemocompatibility o test materials. We have developed a new apparatus by modifying a con and plate-type viscometer to evaluate platelet-biomaterial interactions, th initial event of thrombus formation, in real-time with relatively small platelet suspension volumes (less than 500 μl-PRP/22.4 cm^2-test material) under shear flow conditions. Using this apparatus, microscopic images o platelet adhesion to 14 test materials including transparent, opaque, plate porous, and membranous materials (agarose gel, glass, cellulose, nylon 6 acrylate resin, polyvinyl chloride, nylon 12, silicone-coated-glass, polycarbonate, silicone sheet, polytetrafluoroethylene, and expande polytetrafluoroethylene) under shear flow in real-time were clearl visualized. According the image analyse of adhered platelets, platelet adhesion increased with decreasing contact angle of the materials under 0. dyne/cm^2. On the other hand, under 5.0 and 15 dyne/cm^2, platele adhesion appeared to be maximized on surfaces with intermediat wettability ranging from 60° to 80°. The results showed that th difference of shear stress change the order of hemocompatibility of the test materials. Therefore, it is suggested that shear stress is a necessary component of a valid system for evaluation of anti-thrombogenic materials, and our apparatus should become a useful instrument for testing an screening the biocompatibility and hemocompatibility of biomaterials.

K. Furukawa[1], T. Ushida[1, 2], H. Sugano[3], T. Tamaki[3], N. Ohshima[2], T Tateishi[1, 2]

1. Agency of Industrial Science and Technology, Ministry of Internationa Trade and Industry; 2. University of Tsukuba; 3. Nippon Institute o Technology

QUANTITATION OF CONTACT LENS CONTAMINANTS

Morton D. Prager and Ronald Quintana, UT Southwestern Med. Ctr., Dallas, TX and Alcon Labs, Ft. Worth, TX

An artificial tear (AT) preparation containing 6 proteins and 6 lipids was developed as a deposit model (DM) for contact lens contamination. Quantitation was achieved by labeling with ^{14}C and ^{3}H and measuring the amount bound to the lens. Comparisons were made between deposition from AT and simpler DMs. Lens types included rigid gas permeable (RGP) and the four FDA groups of hydrogel lenses.

A simple DM with lysozyme, a major lens contaminant, as the only protein in an isotonic solution gave deposits of 2.5-4.5 μg on Group I, II, III, and RGP lenses (24 hr) but 125 μg on Group IV (ionic, high water content) lenses. From the more complex AT preparation, lysozyme deposition decreased except for Group IV which showed an increase to 256 μg. Thus interaction of substances in the deposit solution affect deposition of the individual components. Extraction of lenses with acidified aqueous acetonitrile removed >90% lysozyme, but for Group IV lenses 3X as much radioactivity was recovered as was detected by counting the lens directly. This finding was used to estimate that about 1/3 of the lysozyme was at the surface of the Group IV lens while the remainder penetrated the gel network.

Deposition of cholesteryl oleate, dioleoyl phosphatidylcholine, and triolein was examined on hydrogel and RGP lenses. The deposit was always <1 μg/lens. A limitation was the amount of lipid that could be incorporated into AT; e.g., as the concentration of triolein increased from 1.8 to 8.8 μg/ml AT, deposits increased form 3 to 60 ng/lens.

This radiochemical methodology has proved to be a highly sensitive way to measure binding to contact lenses.

USE OF MARS IN THE TREATMENT OF ACUTE LIVER FAILURE. PRELIMINAR MONOCENTRIC EXPERIENCE.

Gilnardo Novelli, MD[1], Massimo Rossi, MD[1], Renzo Pretagostini, MD[1], Massimo Iappelli, MD[1], Luca Poli, MD[1], Fatima Della Pietra, MD[1], Giorgio Della Rocca, MD[2], Pasquale Berloco, MD[1], Aniello Di Nicuolo, MD[1], Daniela Peritore, MD[1], Mara Colonnello, MD[1], Cinzia Cancrini, MD[1], Adolfo Maria Attili, MD[3], Raffaello Cortesini, MD[1]. [1]II Patologia Chirurgica. [2]Ist. di Anestesia e Rianimazione. [3]III Clinica Medica (Cattedra di Gastroenterologia), Università degli Studi "La Sapienza", Roma, Italy.

The bioartificial liver (BAL), with depurative and synthetic capacities, and the molecular absorption recycling system (MARS) caracterized by the only depuration function through hybrid layers foe detoxics tield up to the albumin transportation are two different extracorporeal methods today aviable for the treatment of acute liver failure. The BAL using hepatocites derived from hepatoblastoma or from animals is loaded by the risk of the apparence of antobodies for the exposure to *specific species antigens*, in limitating the application in a longe period. MARS, conditioned from the impossibility of synthesis, can be used for an illimitated time. Aim of our study is to report our experience in using MARS in three patients with acute liver failure secondary to PNF (case 1), FHF due to HBV infection (case 2), Autoimmune acute hepatitis (case 3). All patients were in the waiting list for an urgent liver transplant. The parameters of hepatic and renal function, the coagulation values, *ammoniemia and albuminemia* have been monitorized before and after treatment that lasted 8.1±1.5 hours. The three patients were in hepatic coma and underwent to the EEG. For the lack of hepatic synthesis has always been set up substitutive therapy. Neither hemodinamic complications or varations of the *albuminemia* have been observed during the treatment. Patient nr. 2 has been successfully transplanted. The results of our study are reported in the table I. We conclude that MARS treatment should be safe and that it could be used in hepatic liver failure as bridge to the liver transplantaion or as artificial depurative support waiting for the recovery of normal hepatic function, as it can remove the toxic substances like ammonium and bilirubine.

Case	Age	Sex	Nr. Treat	Alive	Creat. Pre/Post	Bilirubine Pre/Post	INR Pre/Post	Ammonium Pre/Post	Complicati ons
1	40	M	24	Y	0,7/1,2	15/23	2,6/1,8	287/142	None
2	32	F	1	Y	0,7/1,4	17/13	4,1/3	243/109	None
3	45	F	2	N	0,3/0,4	26/23	2,5/2,1	260/70	Infection

FRACTURAL CHARACTERISTICS OF MICROCAPSULE SUSPENSION FOR HEMOLYSIS EVALUATION USING ROTATIONAL SHEAR STRESSER

PURPOSE It is important to evaluate hemolytic properties in the development of circulatory artificial organs. Though animal blood have been used as relative hemolysis evaluation, the results highly depend on individuality. We are developing microcapsule suspension, which is expected to be used for absolute hemolysis evaluation instead of animal blood. In this report, we describe fractural characteristics of microcapsule suspension under fluid dynamical shear stress, which is generated by rotational shear stresser developed in our laboratory.

METHODS We tested two types of microcapsule suspension, MS-100 (100micrometer diameter and polyurethane membrane) and MS-10 (10micrometer diameter and melamine membrane) under the shearing velocity of 0-15,000s-1 for 15min. These fractural characteristics were compared with the results by bovine blood.

RESULTS Fractural properties of MS-100 was similar to that of bovine blood, whereas MS-10 showed different result. This tendency depends on the flow condition in the shear stresser. After clarifying the effect of the flow condition, we expect that fractural characteristics under fluid dynamical shear stress can be simulated utilizing microcapsule suspension. This new method leads to absolute hemolytic evaluation.

O. Maruyama[1)], T. Yamane[1)] M. Nishida[1)], A. Aouidef[1)], T. Tsutsui[2)], T. Jikuya[2)], T. Masuzawa[3)]
1) Mechanical Engineering Laboratory, 2) University of Tsukuba, 3) Ibaraki University

BIOARTIFICIAL TRACHEA PREPARED FROM TRACHEAL ALLOGRAFT: MAINTAINING THE VIABILITY OF CARTILAGE IS NECESSARY FOR IMMUNOSUPPRESSANT-FREE TRACHEAL TRANSPLANTATION

Purpose: To prepare low antigenic tracheal allografts, detergent treatment was done on the trachea for immunosuppressant-free allotransplantation. Animal experiment was performed to evaluate its efficacy. Methods: In group 1(n=5), the fresh canine tracheal segments were treated with a detergent at 4°C for 48 hours and thoroughly rinsed each one with physiological saline solution to remove the detergent. In group 2 (n=6), similar treatment was carried out but at room temperature. In group 3 (n=4), the detergent treatment was performed at 4°C for 48 hours, but the rinsing was omitted The grafts were then used to perform intrathoracic five-ring tracheal replacement in dogs without immunosuppressants. Results: In group 1, the appearance of the chondrocytes in the graft was similar to that in fresh trachea, indicating that the viability of the chondrocytes was maintained. The epithelium and mixed glands had been removed completely, and the grafts were incorporated by the host trachea without stenosis. In group 2 and 3, the epithelium and mixed glands had been removed completely, but the cartilage viability appeared to have been lost. Most of the animals died within one month after transplantation. The cause of death was tracheal stenosis induced by collapse of the graft. Conclusion: These results suggest that maintaining the cartilage viability in the graft is necessary for successful tracheal immunosuppressant-free allotransplantation.

Y. Liu, T. Nakamura, Y. Yamamoto, T. Sekine, H. Ueda, M. Inoue, K. Matsumoto, M. Yoshitani, Y. Hori, T. Toba, Y. Shimizu.
Institute for Frontier Medical Sciences, Kyoto University.

QUANTITATIVE ANALYSIS OF CELL DETACHMENT BY CONE AND PLATE-TYPE RHEOMETER

To realize artificial blood vessels with small diameter, it is very important to find or develop materials that enhance adhesive force of endothelial cells. Therefore, we developed an apparatus to facilitate quantitative analysis of the detachment process of endothelial cells from opaque materials by shear flow. A cone and plate-type viscometer provides a well-defined homogeneous laminar flow, and consists of simple structure. Then, we developed a new apparatus by modifying a cone and plate-type rheometer. To achieve real-time observation of detachment of endothelial cells from either transparent or opaque biomaterials, we combined a modified cone and plate-type rheometer with an upright epi-fluorescence microscope with a transparent cone. To reduce cell number, medium volume, and the material size, we made a small cone with diameter of 10 mm, angle of 20'. To enable fixation of various biomaterials (*i.e.*, plate, membrane and porous-shaped materials) of small diameter (13-16 mm), we improved the plate to which the test biomaterials were fixed so that it was composed from parts: a stainless steel bottom plate and a ring. The mechanical rotation of the cone was guaranteed by three gears connecting directly with the motor. Human umbilical cord vein endothelial cells were labeled with a fluorescent dye which adsorbed to membrane of cells, and inoculated to polycarbonate membrane adsorbed with laminin. After incubation of 24 h, endothelial cells on poycarbonate membrane were exposed to shear stress of 100 dyne/cm^2. The detachment process of endothelial cells was clearly visualized through the transparent cone. When shear stress of 100 dyne/cm^2 was loaded to the surface for 60 min, the ratio retained of initially inoculated cells was 14%. With increase of shear stress, the ratio of the cells retained on the material surface decreased. Furthermore, increase of loading time of shear stress also decreased the ratio of cells retained on the surface. These data suggested that our apparatus can analyze quantitatively detachment of endothelial cells with ease. Therefore, our apparatus should be an useful instrument to develop and evaluate the materials which enhance endothelialization.

K. Furukawa[1], T. Ushida[1, 2], T. Nagase[3], H. Nakamigawa[3], T. Noguchi[3], T. Tamaki[3], T. Tateishi[1, 2]
1.Agency of Industrial Science and Technology, Ministry of International Trade and Industry; 2.University of Tsukuba; 3.Nippon Institute of Technology, Japan.

PROCOAGULANT RESPONSES OF PLATELETS ATTACHED TO BIOMATERIALS

V. Balasubramanian and S.M. Slack, Biomedical Engineering Department, The University of Memphis, Memphis, TN, USA

The plasma protein fibrinogen readily adsorbs to the surface of biomaterials and, because of its demonstrated ability to support platelet adhesion and aggregation, plays a role in thrombotic events associated with the implantation of synthetic materials in the human body. Thus, understanding the factors influencing the interactions of fibrinogen with biomaterials, and how platelet responses are affected, is crucial for the development of synthetic materials exhibiting improved blood compatibility. In this investigation, the effects of protein residence time, co-adsorbed proteins, and shear rate on the procoagulant activity of adherent platelets, along with their morphological status, as deduced from SEM, were studied.

In order to investigate whether adherent platelets promoted the generation of thrombin, samples (polytetrafluroethylene, polyethylene, and silicone rubber) preadsorbed with fibrinogen and exposed to platelets at different wall shear rates were incubated with clotting factors for 5 minutes under static conditions. The amount of thrombin generated per platelet was calculated from the optical density of the color developed by adding substrate S-2238.

SEM images of the platelets revealed that the platelets exhibited different morphologies depending on the shear rate and residence time of the adsorbed fibrinogen. Platelets ranged from their normal discoid shape observed primarily under static conditions to that of fully spread platelets. Results from this study show that platelets, in the presence of shear forces, undergo activation on exposure to surfaces on which adsorbed proteins have resided for short residence times rather than long residence times. Surprisingly, studies determining the procoagulant responses of such adherent platelets demonstrated that the platelets attached to the protein-coated materials did not promote thrombin formation. Very low prothrombinase activity was measured from the adherent platelets, suggesting that the surface-bound fibrinogen primarily potentiates platelet-surface interactions, thus playing a major role in thrombotic complications associated with vascular prosthesis